

Synoptic CB: Porewater DOC, TN

October 2022 Samples

2026-06-17

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##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE
Date_Run = "11/6/2022" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Zoe Read" #Code user
run_notes = "Standard Curve assessed manually on the instrument. Some sample IDs are missing from metadata"
samples <- c("GCrew", "Gcrew", "GWI", "MSM", "SWH") #whatever identifies your samples within the same run

samples_pattern <- paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "_Chk_" #what did you name your check standards?

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name = c("Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202210_1.txt") # , "Raw Data/TOCTN_COMPASS_202210_1.txt"
```

```

#file path and name for raw all peaks file
# raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202210.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2022.csv"

#qac log file path for this year
#Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2022.csv"

```

```
##Set Up Code
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

0.1 Import Data Functions

0.2 Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## * ' -> '...14'
```

```
## # A tibble: 6 x 4
```

```
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>    <dbl> <chr>
## 1 1_202210_Gcrew_TR_LysA_20cm 31.0    1.52 11/6/2022 9:24:35 PM
## 2 2_202210_Gcrew_TR_LysA_45cm 10.1    0.400 11/6/2022 9:46:48 PM
## 3 3_202210_Gcrew_TR_LysB_10cm  5.79    0.319 11/6/2022 10:14:32 PM
## 4 4_202210_Gcrew_TR_LysB_20cm 13.1    1.07 11/6/2022 10:36:35 PM
## 5 5_202210_Gcrew_TR_LysC_10cm 26.4    0.604 11/6/2022 11:06:52 PM
## 6 6_202210_Gcrew_TR_LysC_20cm 55.4    1.96 11/6/2022 11:26:14 PM
```

```
##Fix sample naming
```

```
#remove number from sample ID
```

```
dat_raw <- dat_raw %>%
  mutate(sample_name = str_remove(sample_name, "^[^_]*_"))
```

```
##fix GCW designation
```

```
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("Gcrew", "GCW", sample_name)) %>%
  mutate(sample_name = gsub("Gcrew", "GCW", sample_name))
```

0.3 Assessing Standard Curves - assessed manually on the instrument

0.4 Assess Check Standards

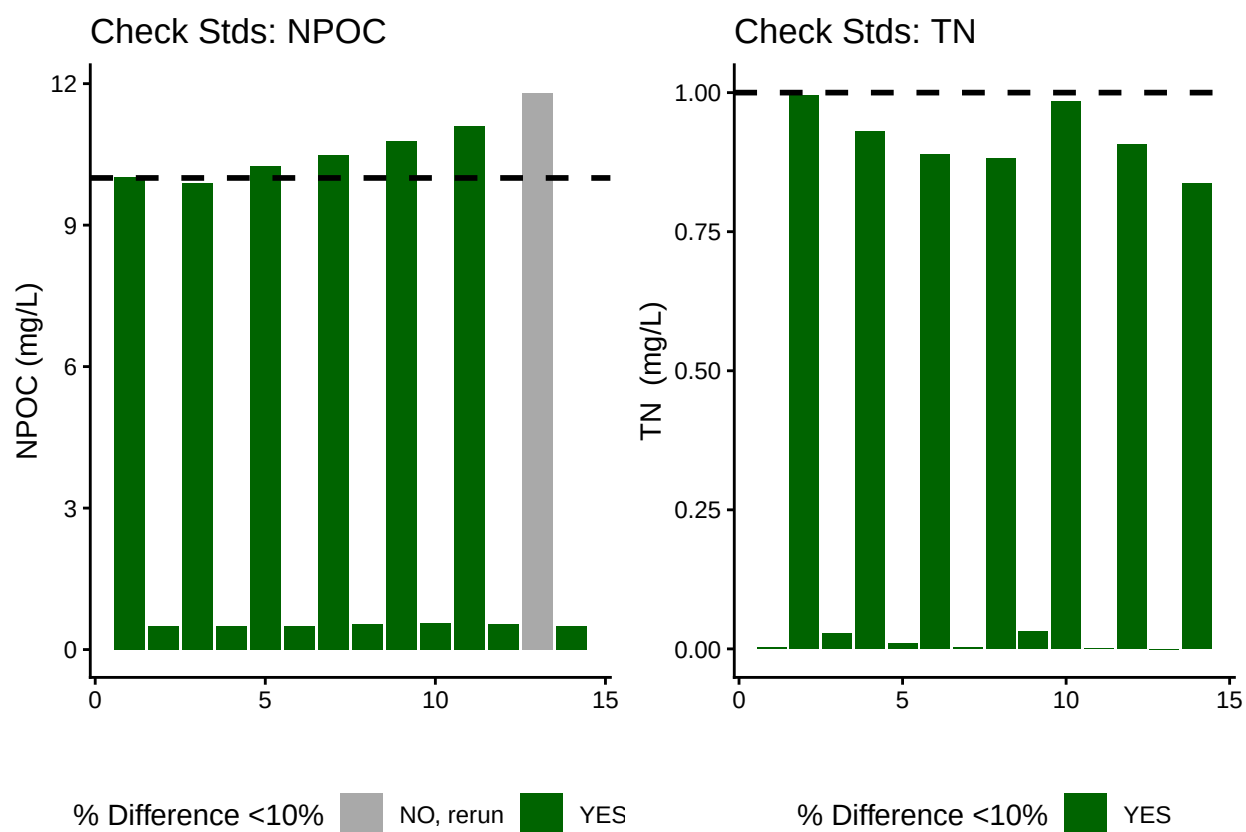
```
## Assess the Check Standards
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">50% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">50% of Nitrogen Check Standards are within range of expected concentration"
```

0.5 Assess Blanks

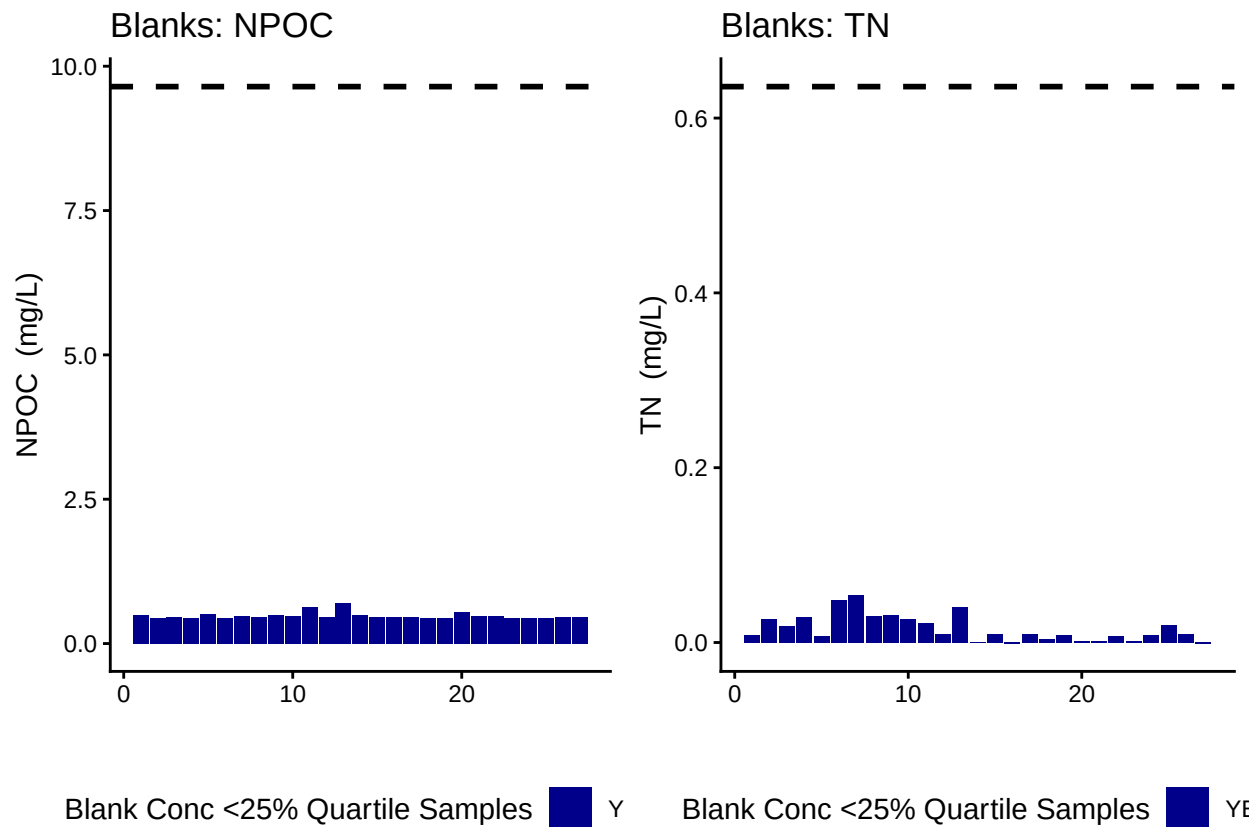
```
## Assess Blanks
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] ">50% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">50% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] 0.4803556
```

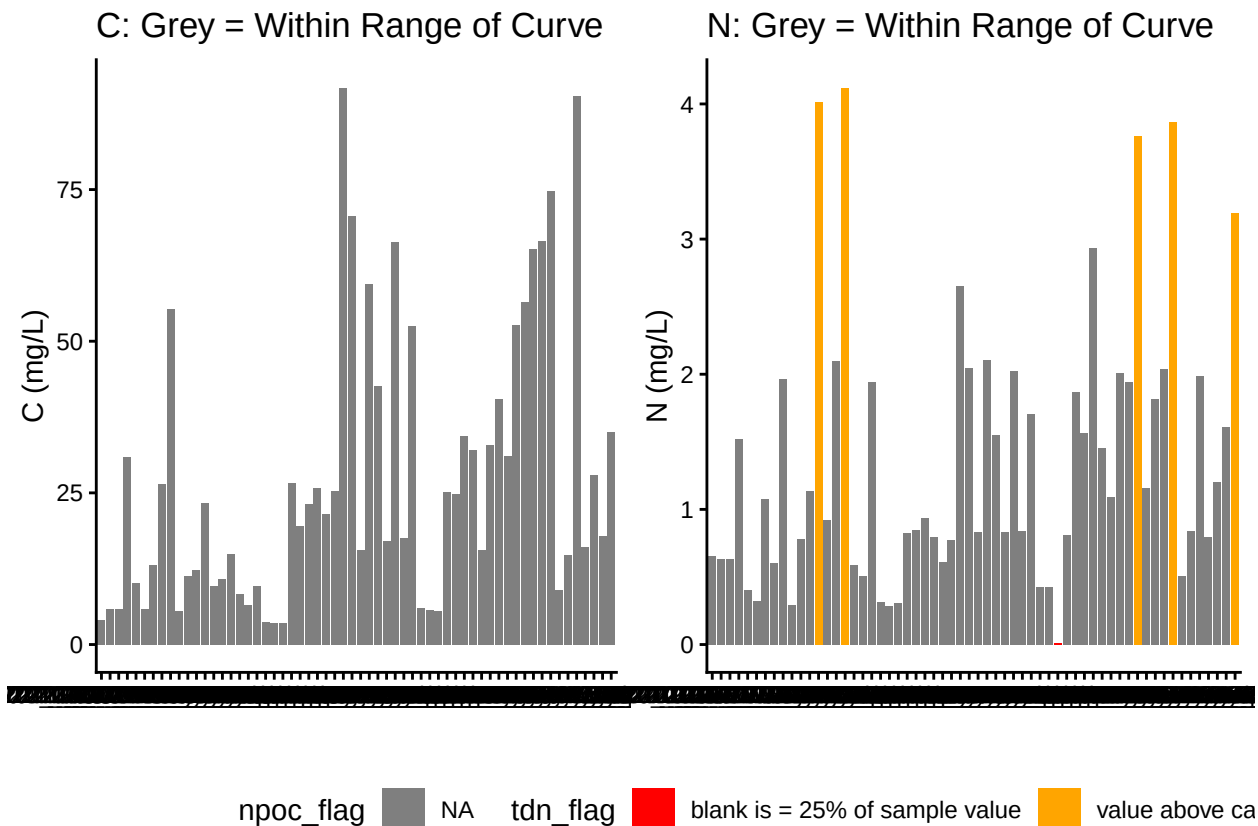
```
## nitrogen blanks:
```

```
## [1] 0.01612222
```

0.6 Assess Duplicates - no duplicates ran

0.7 Sample Flagging - Are samples Within the range of the curve?

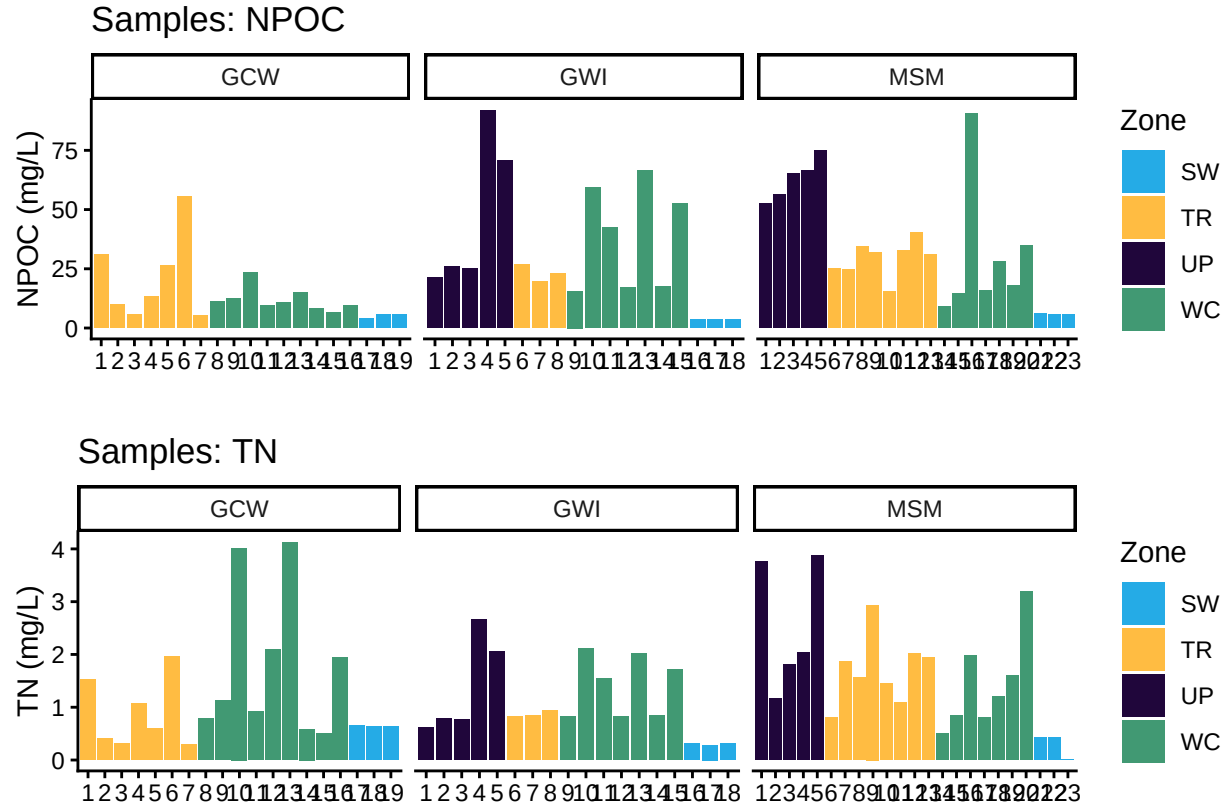
```
## Sample Flagging
```



0.8 Visualize Data by Plot

```
## Visualize Data
```

```
## Warning in rbind(c("202210", "GCW", "TR", "LysA", "20cm"), c("202210", "GCW", :
## number of columns of result is not a multiple of vector length (arg 17)
```



0.9 Convert data from mg/L to uMoles/L

0.10 Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata
```

```
## Some sample IDs are missing from metadata.
```

```
## [1] "202210_GCW_WC_SipA_10cm" "202210_GCW_WC_SipA_20cm"
## [3] "202210_GCW_WC_SipA_45cm" "202210_GCW_WC_SipB_10cm"
## [5] "202210_GCW_WC_SipB_20cm" "202210_GCW_WC_SipB_45cm"
## [7] "202210_GCW_WC_SipC_10cm" "202210_GCW_WC_SipC_20cm"
## [9] "202210_GCW_WC_SipC_45cm" "202210_GWI_UP_LysC_10cm"
## [11] "202210_GWI_UP_LysC_20cm" "202210_GWI_TR_LysB_45cm"
## [13] "202210_GWI_TR_LysC_10cm" "202210_GWI_WC_SipA_10cm"
## [15] "202210_GWI_WC_SipA_20cm" "202210_GWI_WC_SipA_45cm"
## [17] "202210_GWI_WC_SipB_10cm" "202210_GWI_WC_SipB_45cm"
## [19] "202210_GWI_WC_SipC_10cm" "202210_GWI_WC_SipC_20cm"
## [21] "202210_MSM_TR_LysB_45cm" "202210_MSM_WC_SipA_10cm"
## [23] "202210_MSM_WC_SipA_20cm" "202210_MSM_WC_SipA_45cm"
## [25] "202210_MSM_WC_SipB_10cm" "202210_MSM_WC_SipB_20cm"
## [27] "202210_MSM_WC_SipC_10cm" "202210_MSM_WC_SipC_20cm"
```

0.11 Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 21
##   Project      Region Site Zone Replicate Depth_cm Sample_ID Year Month Day
##   <chr>      <chr> <chr> <chr> <chr>      <chr>      <chr>    <int> <int> <int>
## 1 COMPASS: Sy~ CB    GCW  TR    LysA        20      202210_G~  2022    10    11
## 2 COMPASS: Sy~ CB    GCW  TR    LysA        45      202210_G~  2022    10    11
## 3 COMPASS: Sy~ CB    GCW  TR    LysB        10      202210_G~  2022    10    11
## 4 COMPASS: Sy~ CB    GCW  TR    LysB        20      202210_G~  2022    10    11
## 5 COMPASS: Sy~ CB    GCW  TR    LysC        10      202210_G~  2022    10    11
## 6 COMPASS: Sy~ CB    GCW  TR    LysC        20      202210_G~  2022    10    11
## # i 11 more variables: Time <lgl>, Time_Zone <lgl>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>

#end
```